STANDARD FIVE

SEMESTER ONE



Prepared by:L.SIKU

MODULE 1

SCIENCE AND SOCIETY

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TOPIC

THE NATURE OF SCIENCE

OBJECTIVES:

- *Demonstrate the following scientific processes: observing, communicating, comparing, organizing, relating, inferring, predicting, analyzing and applying.*
- *Demonstrate how science is an ongoing process of gathering and evaluating information.*

What is Science?

Science is the study of everything around us.

The Scientific Method/Process

The Scientific method has **five basic steps**.

These are;

- **Observation**
- **Question**
- **Hypothesis**
- **Experiment**
- **Applying results**

What is a Question?

A **question** is what Scientists **ask** when they notice something.

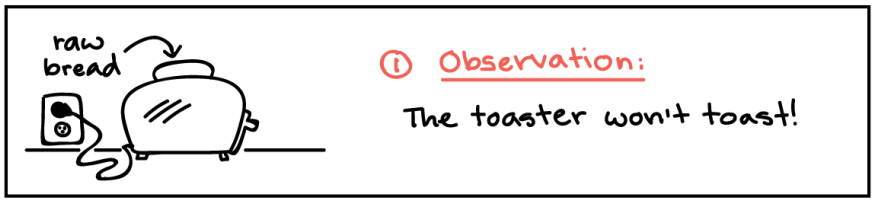
What is a Hypothesis?

A **Hypothesis** is a **possible explanation**

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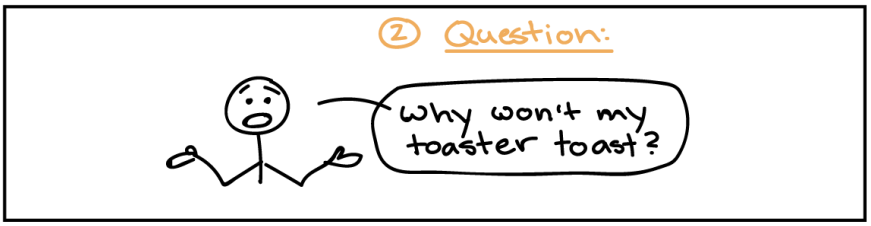
Scientific method example: Failure to toast

1. Make an observation.



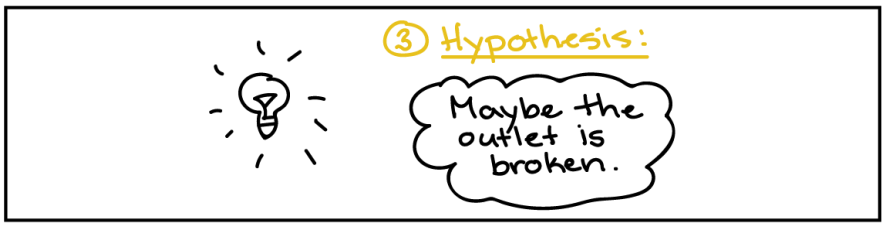
1. Observation: the toaster won't toast.

2. Ask a question.



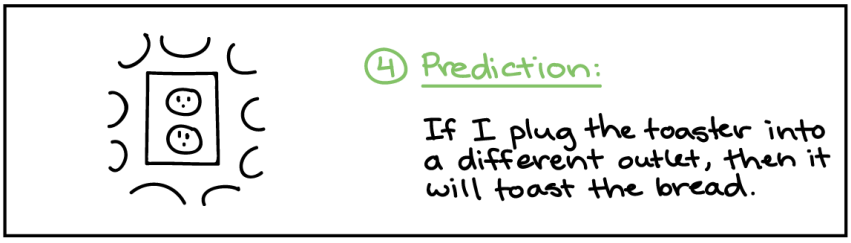
2. Question: Why won't my toaster toast?

3. Propose a hypothesis.



3. Hypothesis: Maybe the outlet is broken.

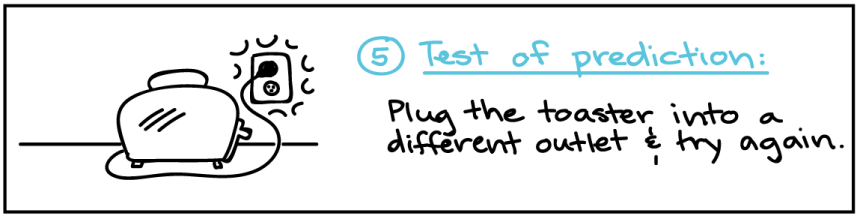
4. Make predictions.



4. Prediction: If I plug the toaster into a different outlet, then it will toast the bread.

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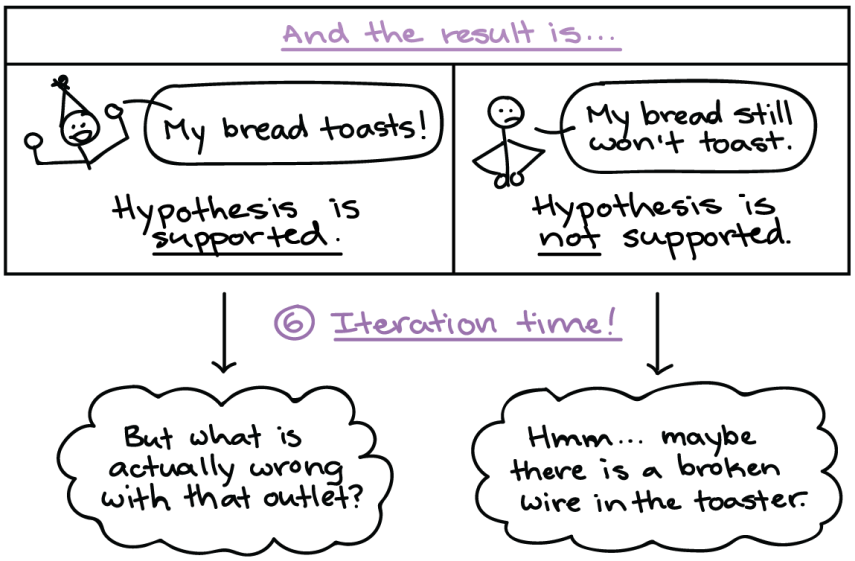
5. Test the predictions.



5. Test of prediction: Plug the toaster into a different outlet and try again.

- If the toaster does toast, then the hypothesis is supported—likely correct.
- If the toaster doesn't toast, then the hypothesis is not supported—likely wrong.

6. Iterate.



And the result is:

Left panel: My bread toasts! Hypothesis is supported. Right panel: My bread still won't toast. Hypothesis is not supported

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QUIZ

1. What is Science?

_____

2. A scientific method has _____ steps.
3. Arrange the following steps of Scientific Methods in order.

   Experiment Question Observation Applying Results Hypothesis

   1. _____
   2. _____
   3. _____
   4. _____
   5. _____

6. Match the following terms with their definitions

Term		Definition
1. Question		**A.** a way for scientists to study and learn things.
2. Scientific Method		**B.** A possible answer to a question
3. Hypothesis		**C.** What scientists ask when they want to find out something.

Read the following paragraph and use it to answer question 7

Mr. Modikwa was going to work in the morning. When he tried to start his car, ***the car would not start.*** He then waited for a minute ***asking himself why the car would not start***. He thought ***maybe it is because the battery is not working***.Mr Modikwa then got out of the car, opened the bonnet and ***removed the car battery and replaced it with a new one***. He then ***tried starting the car again***. This time the car managed to start.***Mr Modikwa then drove his car to work.***

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7. Identify the scientific method steps from the passage above.

Activity	**Scientific method**
Removed the battery and replaced it with another one.	
Asking himself why the car would not start	
Drove his car to work	
The car would not start	
Tried starting the car	

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MODULE 2

NATURE AND THE UNIVERSE

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TOPIC

THE ENVIRONMENT

OBJECTIVES:

- *Describe the interaction among living and non-living components in a given environment.*
- *Explain how the ability of an environment to provide food,water,space and essential nutrients affect its carrying capacity.*

What is the Environment?

The environment is **everything around us.**

Examples:

- **Human beings**
- **Animals**
- **Water**

In the environment there are **living** and **non-living** things.

The carrying capacity

What is carrying capacity?

The **carrying capacity** of an environment means **how many things can live** in an environment **without damaging** it.

Factors that affect carrying capacity

- **Space**
- **Water**
- **Food**

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TOPIC

PLANTS

OBJECTIVES:

- *Investigate conditions necessary for photosynthesis*
- *Explain the importance of photosynthesis to plants and animals*

Plants **make their own food**. They do this through a process called **photosynthesis.**

Photosynthesis takes place in the **leaves of the plant**.

What is needed for photosynthesis to take place.

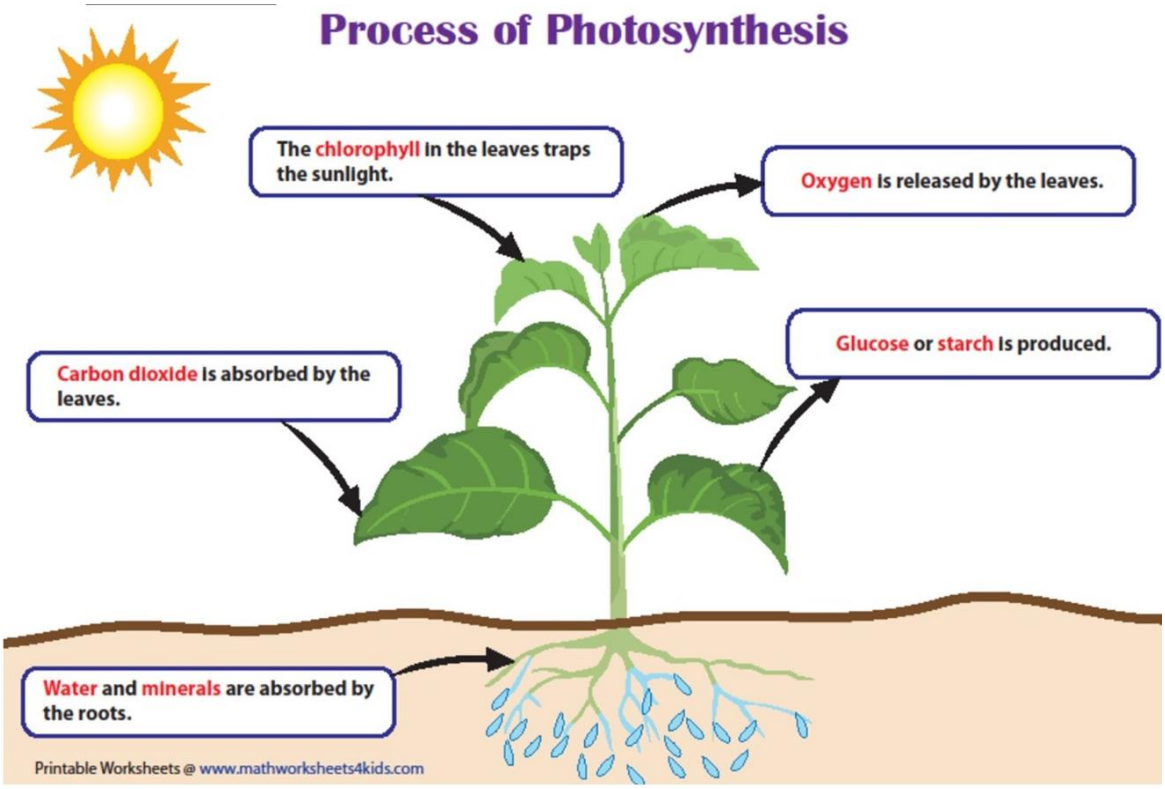
Energy from sunlight

Carbon dioxide from the air.

Water from the soil

Chlorophyl,the green material in leaves.

During photosynthesis the plant releases **Oxygen.**



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QUIZ

Photosynthesis

Multiple choice

1. Which two things combine together in the presence of Sunlight and chlorophyll during photosynthesis to produce food?

   A. Carbon-dioxide and water
   B. Water vapour and manure
   C. Carbohydrates (starch) and Carbon-dioxide
   D. Water and oxygen

2. What do we call the green pigment that is used to capture the energy from the sun during photosynthesis?

   A. Sunlight.
   B. Carbon-dioxide.
   C. Chlorophyll.
   D. Mineral salts.

3. The main function of chlorophyll during photosynthesis is to …….

   A. capture sunlight
   B. produce starch
   C. capture water
   D. store carbon-dioxide

4. The two things that are produced during photosynthesis are

   A. Water and chlorophyll
   B. Starch and Carbon-dioxide
   C. Starch and Oxygen
   D. Chlorophyll and sunlight

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5. What is photosynthesis?

   A. The process whereby plants make their own food.
   B. The washing away of soil
   C. Taking photos using a digital camera
   D. The dying of plants due tom too much sunlight

6. The importance of photosynthesis to animals is that ……

   A. It gives them water
   B. It gives them chlorophyll
   C. It gives them oxygen
   D. It gives them sunlight

7. _____is very important because all people and animals need it to breathe

   A. Oxygen
   B. Water vapour
   C. Carbon-dioxide
   D. Nitrogen

8. During photosynthesis, plants releases…

   A. Nitrogen
   B. Oxygen
   C. Water vapour
   D. Carbon-dioxide

9. The four things that are needed for photosynthesis to take place are:

   A. Water, chlorophyll, carbon-dioxide and oxygen
   B. Water, chlorophyll, carbon-dioxide and Sunlight
   C. Oxygen, water, nitrogen and sunlight
   D. Nitrogen, chlorophyll, water and oxygen

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10. The main function of sunlight during photosynthesis is to ……..

    A. capture sunlight from the sun.
    B. provide energy
    C. capture carbon-dioxide
    D. get water from the roots.

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TOPIC

ANIMALS

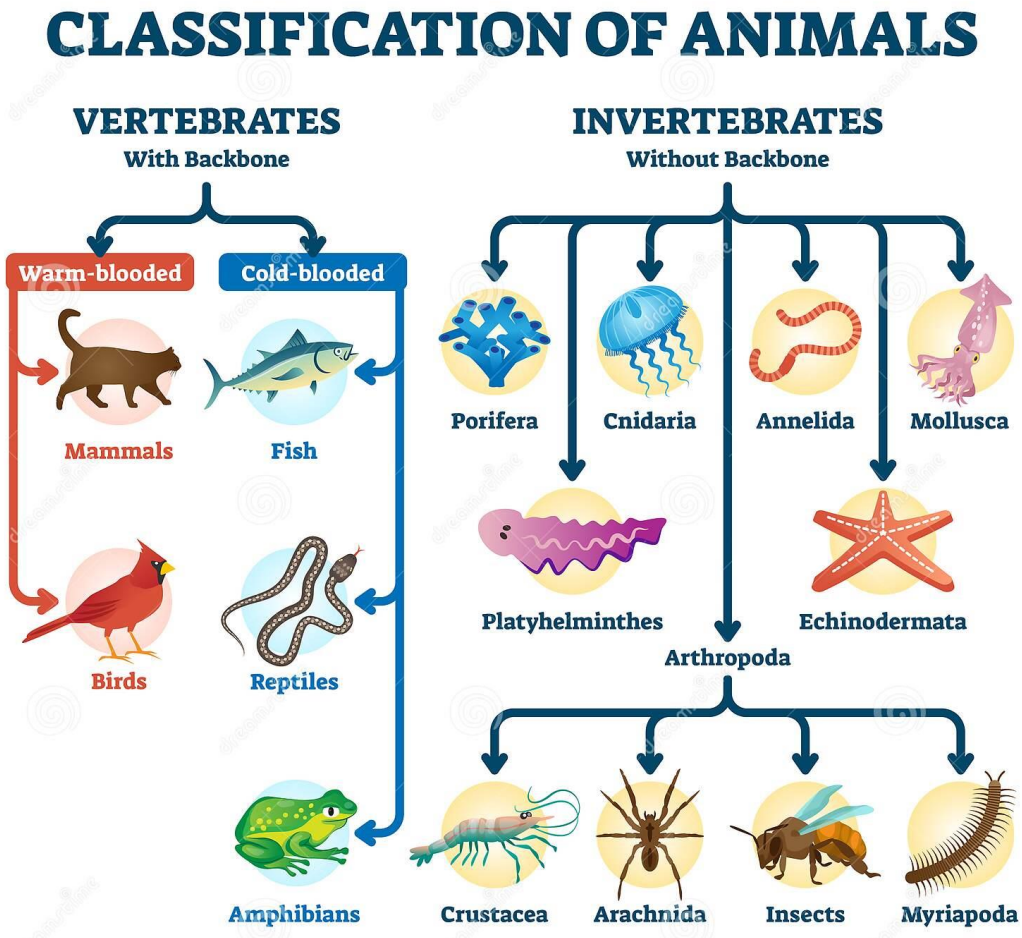
OBJECTIVES:

- *State the characteristics of vertebrate*
- *State the characteristics of invertebrates*
- *Identify classes of vertebrates*
- *Classify animals as vertebrates and invertebrates*

The animal kingdom is divided into **two groups.**

Vertebrates – animals that **have a backbone**.

Invertebrates – animals that **do not have a backbone.**



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Characteristics of vertebrates

- They have a **skeleton** inside their body.It is called an **endoskeleton.**
- Its backbone **can bend** and it **protects the spinal cord** that runs through it.
- It has a **skull** and a **brain.**
- Vertebrates have a **heart** and a **blood stream.**
- All vertebrates have a **pair of eyes.**

Classes of Vertebrates

There are many different animal classes and every animal in the world belongs to one of them. The five most well-known classes of **vertebrates** (animals with backbones) are ***mammals, birds, fish, reptiles, amphibians.***

Characteristics of Vertebrates

	How it Breathes	Body Covering	Born Alive or Eggs	Warm or Cold Blooded	Something that makes them distinctive
Amphibians	Gills as tadpoles, lungs as frogs	Slimy wet skin (moist)	Eggs	Cold	Can live on land and in water
Birds	Lungs	Feathers	Eggs	Warm	Most can fly. All have wings.
Fish	Gills	Scales	Eggs	Cold	Breathe under water. Fins for locomotion.
Mammals	Lungs	Skin and fur (or hair)	Alive. Except the platypus!	Warm	Give birth to live, fully developed young. Suckle young. Mammary glands.
Reptiles	Lungs	Scaly skin	Eggs	Cold	Have scaly skin.

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INVERTEBRATES

An invertebrate is an animal that **does not have a backbone**. Ninety -seven percent of all animal species are invertebrates.

Characteristics

- They **do not have backbone.**
- They are **multicellular**. All cells have different responsibilities in keeping the animal alive.
- Invertebrates are **cold blooded.** They warm their bodies by absorbing heat from their surroundings.
- **Most invertebrates live in water** or spend at least some part of their life in water.
- **Some** groups of invertebrates **live on land.**

Types of Invertebrates

There are many different types of invertebrates and we classify them into groups. The main groups of invertebrates are:

1. Arthropods:
    - Insects.
    - Arachnids.
    - Crustacean.
    - Myriapods.

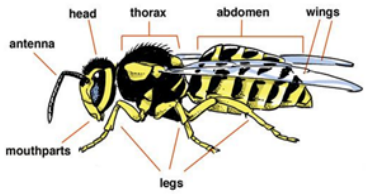
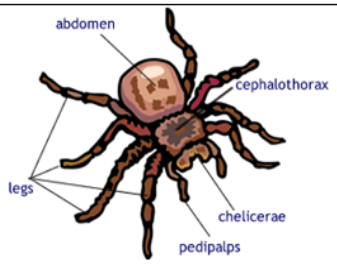
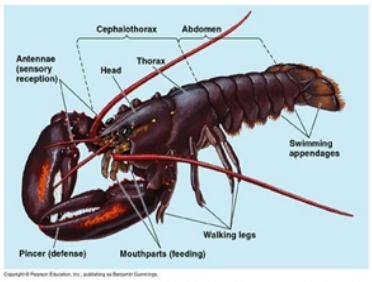
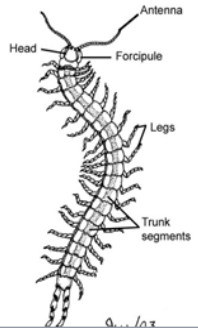
2. Molluscs:
    - Gastropods.
    - Cephalopods.

3. Other invertebrates:
    - Worms.
    - Echinoderms.
    - Jellyfish.
    - Sponges.

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Arthropods

Arthropods are the biggest group of invertebrates. All arthropods have an **external skeleton** that protects their body. They also have **many legs**.

INSECTS	ARACHNIDS	CRUSTACEANS	MYRIAPODS
• **6** legs. • Exoskeleton • 3 body parts (**head, thorax and abdomen**). • Two **antennae**. • Many have **wings**. • Examples are wasps, ants, butterflies…	• **8** legs. • Exoskeleton. • 2 body parts (**cephalothorax and abdomen**). • Most have 8 eyes. • Don´t have antennae or wings • Examples are spiders, scorpions…	• **10** legs. • Exoskeleton. • 2 body parts (**cephalothorax and abdomen**). • Two **antennae**. • Examples are crabs, prawns…	• **Lots** of legs. • Exoskeleton. • Two **antennae**. • Body has lots of segments.
			

Other invertebrate groups

There are other groups of invertebrates that include a variety of different animals. They are:

WORMS	ECHINODERMS	JELLYFISH	SPONGES
• Have a **long soft** segmented **body** that is made of rings. • For example earthworms.	• Have **many small feet**. • For example starfish and sea unchins.	• Can be dangerous. • Have **poisonous tentacles**. • For example jellyfish.	• They cannot move because are fixed to the ground. • Their **body is full of holes**, known as **pores**.
			

Some invertebrates have an **outside skeleton** (also called an **exoskeleton**).

The **largest class** of invertebrates is the **insect** class.

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QUIZ

1. Name the **two groups of animals.**
2. What do we call **animals with a backbone?**
3. Name the **five classes of vertebrates.**
4. Give two **examples of mammals.**
5. Which group of vertebrates are **born with hard-shelled eggs?**
6. Give three **examples of reptiles.**
7. What does the **word amphibian mean?**
8. Which group of vertebrates **have gills,scales and fins?**
9. Where are **amphibians born?**

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TOPIC

AIR

OBJECTIVES:

- *Define air as a mixture of gases*
- *List the major components of air*
- *State properties of air*
- *Demonstrate that a mixture of air and water causes rusting*
- *Demonstrate ways of preventing rusting*

Air

Every empty space is filled with a mixture of gases, called air.

The layer of air round the earth is called atmosphere.

Gasses present in air are nitrogen, carbon dioxide, oxygen,water vapour, helium, argon and many more.

Importance of Air

Without air life cannot survive.

People breathe in oxygen, without which we can't live even for a minute.

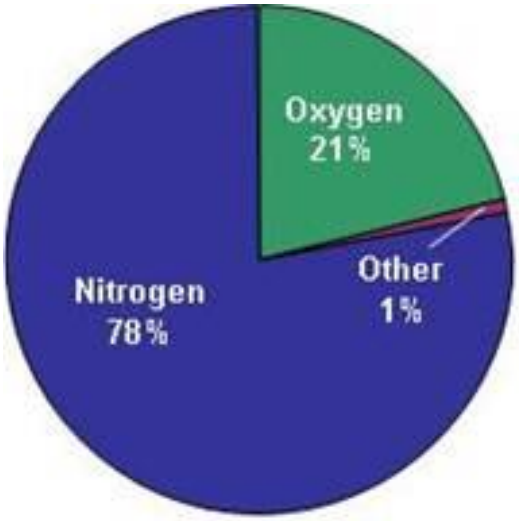
Not only humans, all kinds of life, that is, cats, dogs, microorganisms, birds, snakes, lizards, frogs or even plants need oxygen to survive.

Carbon dioxide is used by plants during photosynthesis to produce their food.

Water vapour is necessary to form clouds that will bring rain.

Air contains 78% nitrogen, 21% of oxygen and 1% is contributed by other gases, including carbon dioxide.

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Exercise

1. If we keep all the components of air in increasing order according to their percentage, which one of the following is correct?

(a) Carbon dioxide. Nitrogen, oxygen

(b) Nitrogen, Oxygen, Carbon dioxide

(c) Carbon dioxide, oxygen, nitrogen

(d) Oxygen, carbon dioxide, nitrogen

(e) None of these

2. The thick blanket of air, which surround the earth, is called

(a) Atmosphere

(b) Biosphere

(c) Hydrosphere

(d) All of these

(e) None of these

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3. Water vapor helps to form

(a) Atmosphere

(b) Oxygen

(c) Clouds

(d) Sun

(e) All of the above

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Properties of Air

Air takes up space

Air has mass/weight

Air is affected by heat

Air exerts pressure

Air can be compressed

Air Takes Up Space

The balloon expands because you are putting something into the balloon; air. This air takes up space, so the more air you put into the balloon, the more space it takes up.

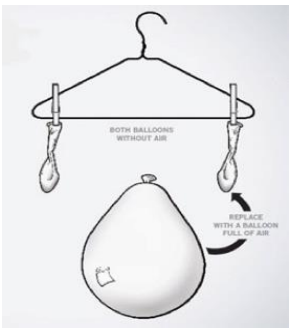
When you use a pump to blow up a football, you don't put nothing into it, you put air into it - this air takes up space which is why the football expands.



Air Has Mass

Make a balance with a stick or coat-hanger suspended by a string in the middle. Tie an empty balloon on each side to prove they weigh the same. Inflate one balloon and rehang it. That side of the balance will be heavier. If air had no mass, there would've been no change.

Air is really quite heavy.

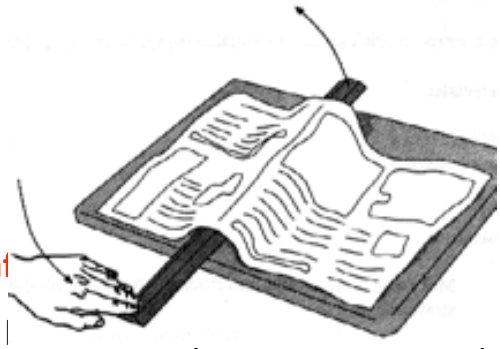


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Air Exerts Pressure

Take a metre stick and lay it on a table. Unfold a full page of a newspaper and lay it flat over the metre stick. Push down on the other end of the metre stick. What happens? Why can't you lift a super-light piece of paper? **Air exerts pressure (in all directions)**.

The air above the paper pushes down on it (pressure). This pressure is what makes the paper lay flat on the table - it's being pushed down.

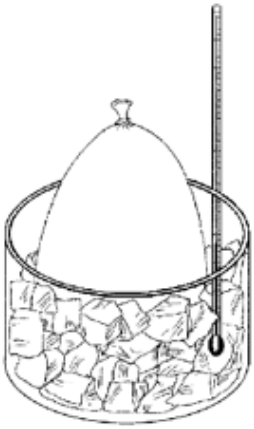


Air is a

Take a . of a pop bottle. Observe the size of the balloon now (at room temperature). Now place it in a freezer for 10 minutes, remove it and observe size of balloon. Now take the bottle and hold it in a baking dish of almost-boiling water for 10 minutes. Now let the bottle sit on the table for 10 minutes. You should now see the balloon return to the same size as it was to start.

The greater the temperature, the faster the air particles move (increasing pressure).

The colder the air becomes though, the slower the air particles move (lowering pressure).



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Air can be Compressed

Take a plastic pop bottle and with the cap off the bottle, hold your hand above the mouth of the bottle and squeeze. What do you feel? Screw the cap on tightly and squeeze again. What happens when you squeeze the bottle now? Now, fill the bottle completely with water, replace the cap and squeeze again. What do you feel now?

When you squeezed the open bottle, you forced some of the air out of the mouth. When you placed the cap on the bottle and squeezed again, there was no place for the air to go, but you were able to squeeze the bottle together. In other words, you were able to compress (or squeeze together) the air inside the bottle. However, when you filled the bottle with water and capped it, you could not squeeze the bottle very much at all because you could not compress the water inside.

Gases such as air can be compressed, but liquids such as water, cannot be compressed.



The inflated balloon has compressed air pressing equally against all the sides.



The air rushes out the open hole at the bottom. The action is that the air is pushed out in one direction. The reaction is that the balloon flies in the other direction.

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AIR AND RUSTING

What is rust?

Rust is a reddish-brown or reddish-yellow color that coats iron or steel when exposed to air and moisture.

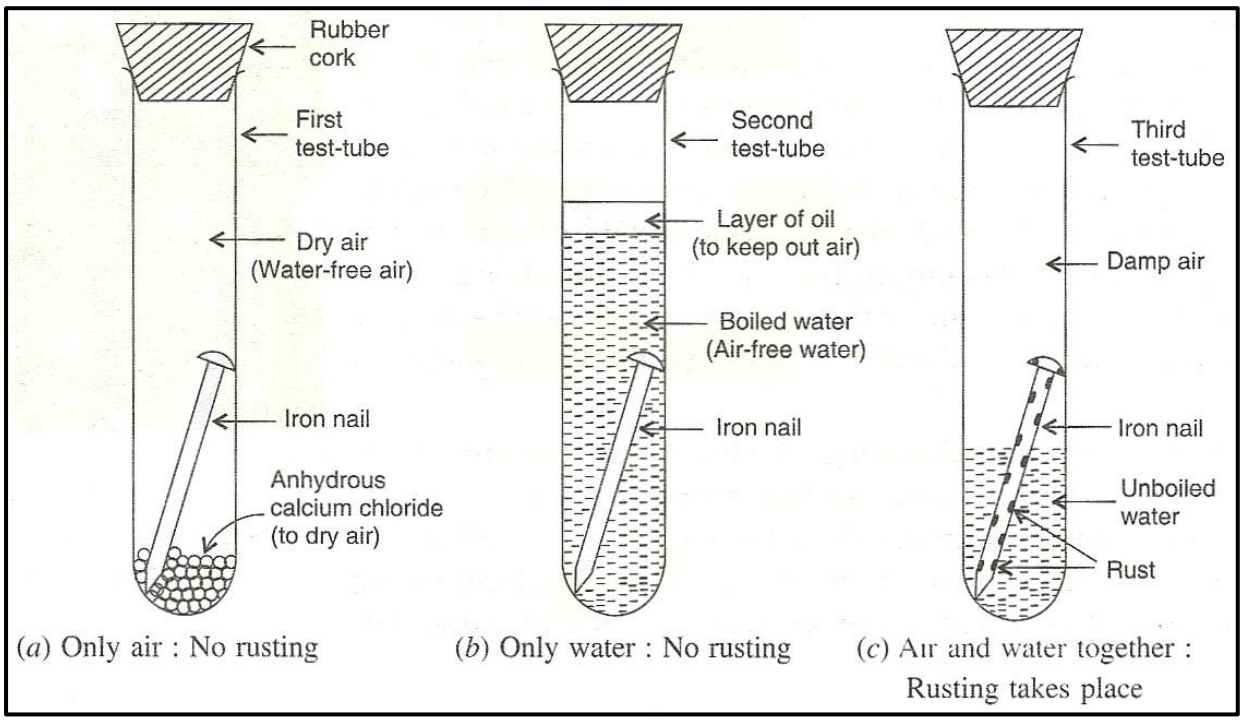
Rusting occurs when iron is exposed to oxygen in air and water or water vapor. ... This process of rust formation in presence of oxygen and water or water vapor is called **rusting**.

Examples of rusting





EXPERIMENT FOR RUSTING



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This experiment shows that;

1. Air alone cannot cause iron and other metals to rust.
1. Iron and other metals will not rust if water (moisture) and air are kept away from the surface of the metal (if covered with oil or grease).
2. Air and water together will cause iron and other metals to rust.

Here are some ways where we can prevent rusting:

1. Paint it - Paint the metal with any durable acrylic paint. ...
2. Keep moisture at Bay - Avoid contact with water or moisture. ...
3. Coat it with Oil - Use of oil or grease significantly reduces the chances of developing rust in iron.

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TOPIC

WATER

OBJECTIVES:

- *Discussthe distribution of water on the earth surface,underground and in the atmosphere*
- *Explain the water cycle*
- *Discuss ways of collecting rainwater*
- *Suggest ways of conserving water*
- *Discuss the social and economic effects of limited water on the lives of people.*

Water is one of the most important substances on planet Earth. All forms of life need water to live. Water is constantly moving about the Earth through the water cycle.

Properties of Water

Pure water actually has no taste at all. It is tasteless.

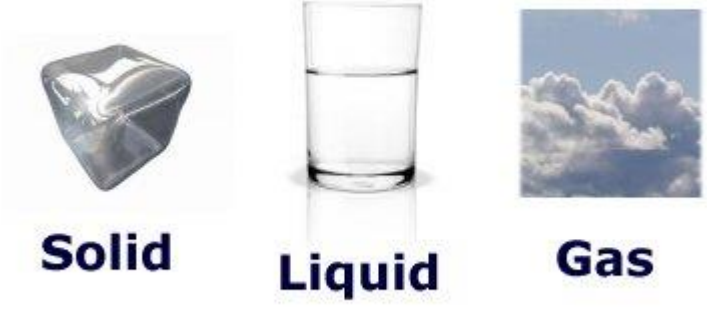
Water has no colour.It is colourless.

Water has no smell. It is odourless.

States of Water

Water can exist in three states:

- **Ice** - Ice is the solid form of water. When water gets below 0 degrees C it will freeze and become ice.
- **Liquid** - This is the wet stuff we drink and swim around in.
- **Vapor** - When water evaporates or gets above 100 degrees C and starts boiling, water turns into its gas state called vapor.



Prepared b

When solid ice is heated, it **melts** and changes to liquid **water**. When the liquid water is heated it **evaporates** and changes to an invisible **gas** called **water vapour.**

Water on Earth

Water is literally all over the Earth. Around 70% of the Earth's surface is covered in water. Most of the water on earth is found on the **surface**. Some is also found **underground.**

Surface water

- Oceans
- Rivers
- Dams

Around 97% of the Earth's water is in the oceans.

Underground water

- Borehole

Water in the atmosphere

- Water vapor

Uses of Water

We use water for more things than just drinking. We use it in agriculture to grow food, in cleaning to wash our clothes and dishes, in firefighting to put out fires, and in hydropower to create energy with dams.

Collecting Rain water

Rainwater can be collected in the following ways;

- Dams
- Reservoirs
- Tanks

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Conserving water

- Fix leaking taps
- Take a shower instead of bath
- Water plants late afternoon
- Use water for more than one purpose
- Do not wash hands or brush teeth under a running tap

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QUIZ

Answer all questions by filling in the following words

underground*, *ice*, *fixing leaking taps*, *melts*, *tasteless*, *dams*, *vapour*, *surface*, *oceans*, *liquid*, *rivers*, *watering plants late afternoon*, *reservoir*, *water*, *tanks*, *colorless*, *evaporates*, *dams*, *odourless*, *boreholes.

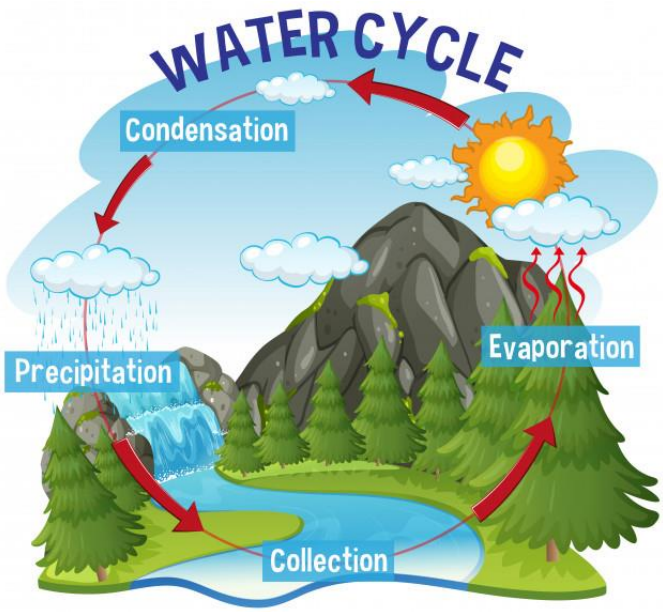
1. All forms of life need _____ to live.
2. Pure water actually has no taste at all. It is_____.
3. Water has no colour.It is _____.
4. Water has no smell. It is _____.
5. _____is the solid form of water. When water gets below 0 degrees C it will freeze and become ice.
6. _____ is the wet stuff we drink and swim around in.
7. When water evaporates or gets above 100 degrees C and starts boiling, water turns into its gas state called _____.
8. When solid ice is heated, it _____ and changes to liquid water.
9. When the liquid water is heated it _____and changes to an invisible gas called water vapour.
10. Most of the water on earth is found on the _____.
11. Some is also found _____.
12. Surface water is found in _____, _____ and _____.
13. Underground water can be found in _____.
14. Rainwater can be collected in the following ways; _____, _____ and _____.
15. We can conserve water by:

_____

_____

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The water Cycle



Evaporation:

Heat from the Sun causes water on Earth (in oceans, lakes etc) to evaporate (turn from liquid into gas) and rise into the sky. This water vapor collects in the sky in the form of clouds.

Condensation:

As water vapor in the clouds cools down it becomes water again, this process is called condensation.

Precipitation:

Water falls from the sky in the form of rain, snow, hail, or sleet, this process is called precipitation.

Collection:

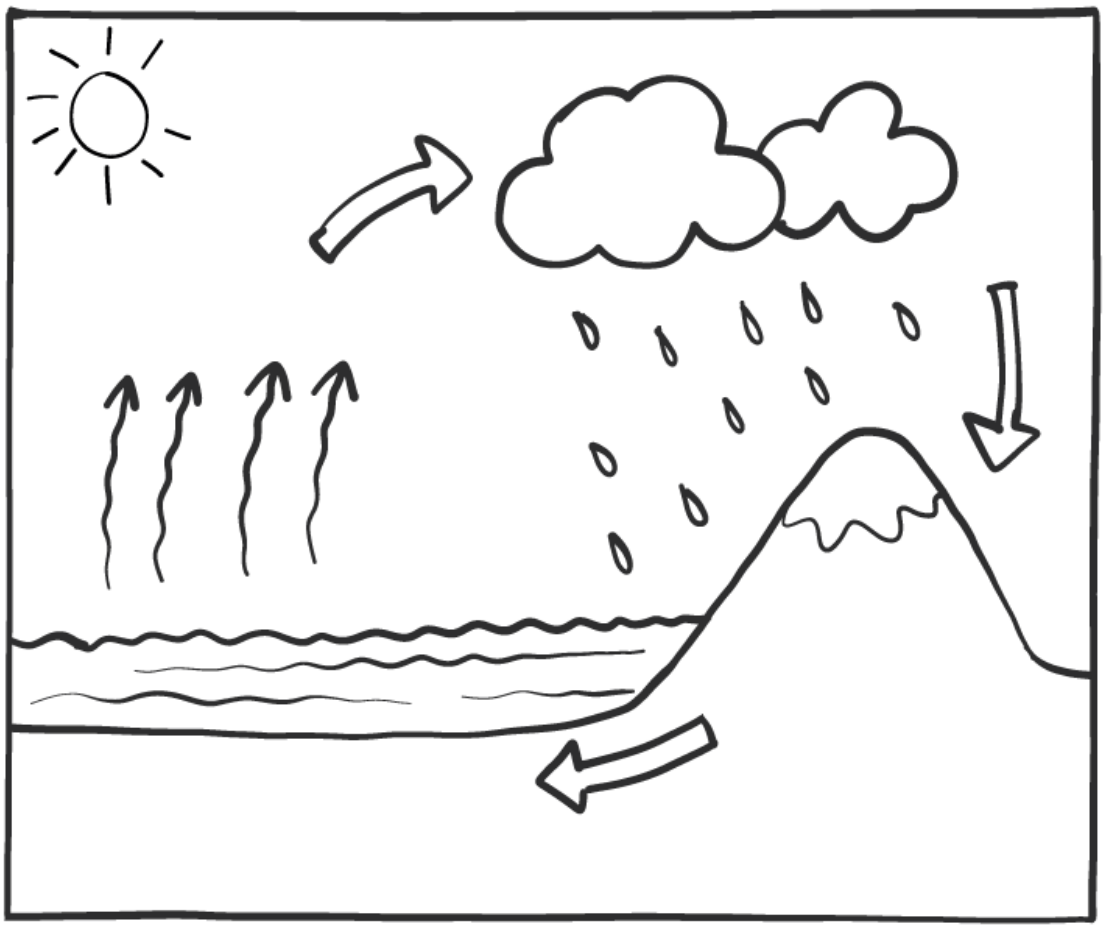
Oceans and lakes collect water that has fallen. Water evaporates into the sky again and the cycle continues.

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QUIZ

The Water Cycle

Color and complete the cycle with these words:
Precipitation, Collection, Evaporation and Condensation



	Rain, snow and sleet from rain clouds
	Water running from land into the ocean
	Sun heats the waters and dries it up
	Water vapor in the atmosphere turns to water

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TOPIC

WEATHER

OBJECTIVES:

- *Construct a simple weather station*
- *Measure weather conditions*
- *Interpret changes in weather conditions*
- *Interpret symbols used on a weather chart*
- *Record weather conditions using conventional symbols*

What is weather?

Weather is the daily condition of the atmosphere recorded over a short period of time.

People who study weather are known as **Meteorologists.**

The study of weather is called **Meteorology.**

Fortelling how the weather will look like is called **Weather Forecast** or **Weather prediction.**

Weather instruments are stored in a **Stevenson Screen.**This is a wooden box with percolated holes to allow free movement of air or free circulation of air.

Weather conditions

Weather conditions can be described as;

- **Windy**
- **Sunny**
- **Rainy**
- **Cloudy**
-

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Elements of weather

Elements of weather are things which make up weather.There are different elements of weather such as;

- Temperature
- Rainfall
- Air pressure
- Humidity
- Wind direction
- Wind speed
- Cloud cover
- Sunshine

ELEMENT	INSTRUMENT	UNIT OF MEASUREMENT
Temperature	 Themometer	Degrees Celsius °C
Rainfall	 Raingauge	Milli metres mm

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Air pressure	 Aneroid Barometer	Milli Bars Mb Or Pascal KPa
Humidity	 Hygrometer	Percentages %
Wind direction	 Wind Vane  Wind Sock	Cardinal Points
Wind speed	 Anemometer	Kilometers per hour Km/Hr
Cloud cover	Observation	Oktas
Sunshine	 Campbell-Stoke Sunshine Recorder	Hours Hrs

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TOPIC

THE SOLAR SYSTEM

OBJECTIVES:

- *Define the universe*
- *Describe common bodies in the solar system*
- *Identify the sun as the nearest star to the earth*
- *Explain why the earth is the only planet capable of supporting life.*

THE SUN



The sun is a **star**.

It is the **nearest star** to the **earth.**

The Sun is about **150 million kilometers** away from the Earth.

The Sun is **5,500 degrees Celsius**.

The Sun is **74 times hotter** than the highest ever temperature on Earth

The Sun gives life to Earth from all the energy the Sun makes

Around **74% of the Sun's mass** is made up of **hydrogen,** which is amazing as it is a very, very light gas, in fact the lightest gas ever found. **Helium** makes up around **24%.**

The rest of the Sun is made up of other heavier gases like **oxygen, carbon, iron** and **neon.**

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LIFE ON OTHER PLANETS

A planet needs three things for organisms to live;

- An **atmosphere** which can be breathed
- A suitable **temperature**
- **Water**

The earth is the only planet in the solar system where life can exist.

It has a suitable temperature.

The earth's atmosphere contains **oxygen** for use by people and animals

It contains **carbon dioxide** for use by plants to make food.

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QUIZ

WEATHER AND SOLAR SYSTEM

1. How many planets are in our solar system?
   A. 4
   B. 6
   C. 8
   D. 10
2. The condition of the atmosphere in a certain place at a certain time is known as _____.
   A. Atmosphere
   B. Weather
   C. Water cycle
   D. Matter
3. What two planets have visible rings around them?
   A. Saturn and Mercury
   B. Saturn and Mars
   C. Saturn and Uranus
   D. Saturn and Earth
4. The air that surrounds the earth is the
   A. Climate
   B. Weather
   C. Matter
   D. Atmosphere

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5. Which planet is the smallest?

A. Neptune

B. Saturn

C. Mercury

D. Uranus

6. _____ is the third planet from the Sun.

A. Mars

B. Earth

C. Venus

D. Neptune

7. A _____ is an instrument that measures rainfall

A. Hygrometer

B. Anemometer

C. Barometer

D. Rain gauge

8. Which is the hottest planet in our solar system?

A. Neptune

B. Mars

C. Saturn

D. Venus

9. The Sun is responsible for _____ percent of the total solar system mass.

A. 99.86

B. 100

C. 50

D. 25

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10. What weather tool is used to measure how hot or how cold something is?

A. Boiling Point
B. Air Pressure
C. Thermometer
D. Temperature

11. Which is the largest planet in our solar system?

A. Jupiter
B. Neptune
C. Mars
D. Venus

12. The anemometer and weather vane are weather tools used to measure wind

A. speed and temperature.
B. direction and pressure.
C. pressure and temperature.
D. direction and speed.

13. How many moons does Mars have?

A. 5
B. 10
C. 20
D. 2

14. This weather instrument measures air pressure

A. Wind vane
B. Barometer
C. Rain gauge
D. Anemometer

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15. Which planet is the furthest from the sun?

A. Earth
B. Mars
C. Neptune
D. Saturn

16. How many days does it take for Earth to make one revolution around the Sun?

A. 31 days
B. 1 day
C. 250 days
D. 365 days

17. Which planet is the closest to the sun?

A. Mars
B. Earth
C. Venus
D. Mercury

18. Inner Planets are

A. Far away from the sun
B. Next to the sun
C. Near the sun
D. Behind the sun

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MODULE 3

MATTER AND ENERGY

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TOPIC

MATTER

OBJECTIVES:

- *Describe the three states of matter in terms of volume and shape.*
- *Classify solids as metals or non-metals*
- *Relate the physical properties of common metals to their uses*
- *Relate the physical properties of non-metals to their uses.*
- *List common household chemicals.*
- *Classify household chemicals into solids,liquids or gases*
- *State the uses of given household chemicals.*

Matter is everything that we come across in our lives, like the air you breathe, the clothes you wear, cool drinks – literally everything!

Matter is defined as anything that **has mass** and **takes up space.**

Matter is found in **3 major states**; ***solid, liquid*** and ***gas.***

There are also ones that are less common and they're called **plasma** and **beam**.

All matter is made of **atoms**. Atoms are the **smallest particle of matter**.

The three states of matter.

Solids

Solids are objects that **keep their own shape** and **do not flow in a given temperature.**

Solids are composed of **atoms packed tightly together.**

Solids can be **different colors** and **textures**, and they can be turned into **different shapes**

Liquids

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Liquids do not **hold their shape at room temperature**.

They **take the shape of the container** they are in and they can flow at a given temperature.

There is **space between the atoms** of a liquid and they move slightly all of the time.

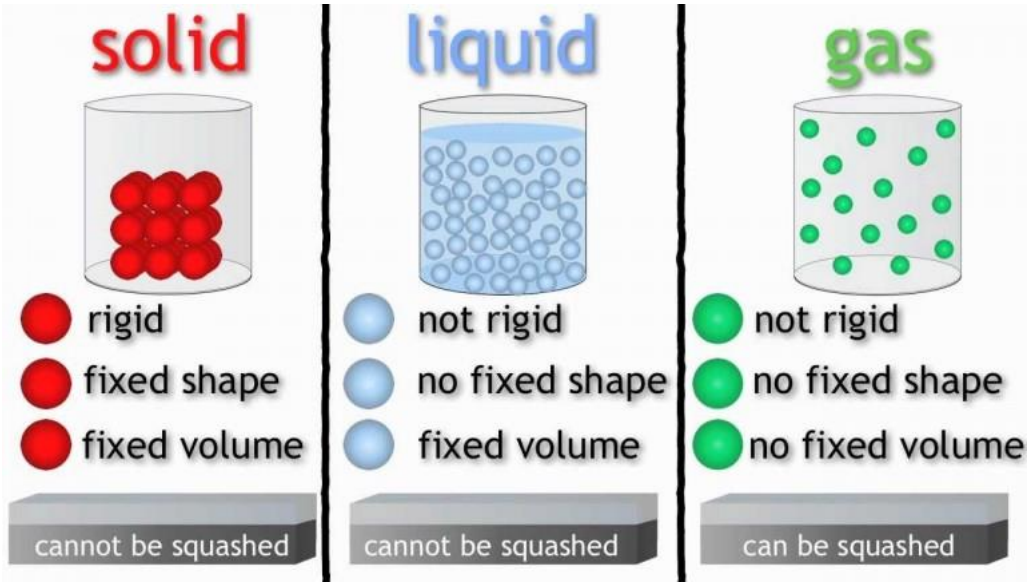
Examples of liquids are ***tea***, ***water*** and ***blood.***

Gases

Gases are air-like substances that can move around freely or flow to fit a container and they don't have their own shape. You can put your hand through gases and you won't feel them.

Gases are always moving. There is so much space between the atoms in gas that you can move around in them easily.

Gases will take on the shape of their container and can be **compressed** into a smaller space.



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Change of State

Matter can move from one state to another, but can still be the same substance. A change of state, also called a **phase change,** is a physical change from one state of matter to another, for example, from solid to liquid or from liquid to gas.

By **lowering the temperature** of water, it can **freeze** into a **solid.** By **heating** water, it can become **steam** which is a **gas**. Whether solid, liquid or gas - water is still water.

Metals and Non-Metals

Solids can be divided into two groups; **metals** and **non-metals.**

Metals are materials holding or possessing the characteristics of being shiny, hard, fusible, malleable, ductile, etc. Except for mercury, metals are solids at room temperature and pressure.

Non – metals are materials not holding the characteristics of metals, means they are not shiny, hard, fusible, malleable, ductile, etc. Nonmetals are typically poor electrical and thermal conductors and don't have a metallic luster. They can be found as solids, liquids, or gases under ordinary conditions.

Examples;

Metals	Non metals
• Iron • Uranium • Sodium • Aluminum • Calcium • Gold	• Nitrogen • Oxygen • Helium • Sulfur • Chlorine • Carbon

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• Silver • Copper • mercury	• Phosphorus • Iodine • Diamond • Graphite

The properties of metals and non-metals

Description of a property	Metals	Non-metals
Colour	Most are **shiny**	Most are **dull**
Hardness	Most are **hard**	Most are **soft**
Strength	Most are **strong**	Most are **weak**
Bending	Nearly all **can bend**	Most **cannot**
Density	**High density**	Mostly **low density**
Hard to melt	Mostly yes	Mostly **easy to melt**
Make a ringing	Mostly yes	No
Conduct heat well	Yes	No
Conduct electricity	Yes	No(except carbon)

Uses of metals

- build things that carry heavy weights
- used for jewellery
- shiny parts of motorcars
- used to make knives or sharp ends of tools for digging
- making wire
- used for electric cables

Uses of non-metals

- (Coal and wood)Used to produce heat.
- (Phosphorus and Sulphur)Used to make heads of matches.
- (Diamond)Used to make drills that can cut through the hardest parts

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STATES OF MATTER QUIZ

1. The particles of matter making up a _____are packed together tightly.
A. piece
B. liquid
C. gas
D. solid

2. Three states of matter are _____.
A. solid, liquid, and gas
B. buoyancy, mass, shape
C. gas, liquid, and mixture
D. density, weight, and gas

3. Which state of matter is easily compressed and easily changes shape to fill its container?
A. solid
B. liquid
C. gas
D. plasma

4. To change from a solid into a liquid is to _____.
A. fluid
B. solidify
C. melt
D. freeze

5. Water is in a _____ state in the form of ice.
A. solid
B. liquid
C. gas

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